

Grade: /15

AI/ML Capstone Project

D2 Project Design Document (15% of final mark)

Due: Week 4 (softcopy on SLATE, hardcopy to the professor in class)

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Instructions:

- Please work on the following deliverable with the group.
- Prepare the deliverable report as discussed in class.
- Type the report using word processing software (preferably using LaTeX, otherwise MS Word is fine).
- Upload the report in PDF format on SLATE. The team leader should upload this deliverable on SLATE as a single PDF file.
- Present a printout (stapled) of the deliverable to the professor in class in Week 4.

Overall Objectives:

In this D2 Project Design Document, the objective is to design a realistic and achievable project plan that defines technical architecture, milestones, and team deliverables using industry standard practices. This document will serve as a foundational blueprint for all subsequent development phases, and it must be structured with clarity and professionalism.

Q1: Project Objectives

Create a Work Breakdown Structure (WBS) detailing technical tasks and deliverables. As part of the WBS, include at least four Work Packages (WPs). These WPs should address the major objectives of the AI/ML system architecture and data flow intended for this capstone project. In the space below, provide an explicit heading for each WP followed by a brief description detailing what exactly will be accomplished (in 4 to 6 sentences). [2 marks]

Work Package 1: Multimodal Data Ingestion and Pre-processing Our goal is to build an automated data pipeline that continuously collects diverse inputs for live traffic safety monitoring. We will write ingestion scripts to pull live text alerts directly from the Ontario 511 API and real-time weather. At the same time, the system will download live image feeds from Ministry of Transportation highway cameras alongside historical collision logs from the Toronto Police. To process the unstructured text, we will apply natural language processing techniques like tokenization to strip out the important words and isolate real road hazards. For the camera feeds, an automated OpenCV workflow will crop the frames and blur sensitive details like license plates or faces to maintain strict privacy standards.

Work Package 2: Predictive Machine Learning Model Development This phase is focused on building and training the intelligent core models that power our multi-layered risk analysis. We will design a Convolutional Neural Network using PyTorch to analyze highway camera frames and classify road conditions as dry, wet, snowy, or icy. For the incoming text data, a TF-IDF vectorizer will mathematically weight the severity of the language found in emergency alerts. We will then build a central Lasso-regularized Logistic Regression model to fuse these visual and text scores with live weather telemetry. Crucially, we will tune our training process to heavily maximize model recall, so the system prioritizes catching dangerous conditions over accuracy.

Work Package 3: Automate Backend Operations and Containerization Goal here is to convert our machine learning scripts into a fully automated backend computing system. We will start by establishing a virtual environment with pinned dependencies, so the codebase runs identically on every team member's machine using GitHub to collaborate. Next, we will implement background automation routines that automatically fetch raw API data and run model predictions every few minutes. We will also serialize our trained neural network weights and configuration files so the backend can instantly load them during live cycles. Finally, the entire application runtime will be inside lightweight Docker containers to guarantee seamless deployment across different computer operating systems.

Work Package 4: Interactive Frontend Application UI The final phase is to build

an intuitive web application that turns complex backend predictions into clear safety insights for everyday drivers. We will begin using existing google map user interface to strategically arrange our warning cards, route suggestion, and map elements. Using the Dash library framework, we will develop the live dashboard by binding our Python prediction endpoints directly to interactive front-facing sliders and buttons. The centerpiece of the interface will be a dynamic geographic map that automatically plots the user's intended driving path. This map module will actively update the color formatting of highway lines based on the specific hazard indices generated by our data fusion models.

Q2: Project Tasks

The focus here is on identifying the granularity of the WPs. For each WP, develop at least 3 to 4 tasks detailing the various subcomponents related to the AI/ML pipeline and applications. In the space below, provide a descriptive heading for each task. Then, for each labelled task, provide an explanation of what the task entails (in 1 to 2 sentences). Avoid being too redundant to what was indicated in Q1. [2 marks]

Work Package 1: Data Gathering & Preparation

- Connecting to Live Data: We will write scripts to connect our system directly to the Ontario511 and weather APIs, ensuring our data feed updates automatically.
- Organizing Past Records: We will gather and clean the historical collision data from the Toronto Police, formatting it so our predictive models can easily learn from past trends.
- Preparing Text: We will write text-processing rules to strip out filler words from emergency alerts, keeping only the critical terms that describe road hazards.
- Blurring Privacy Info: We will set up an automated masking tool that blurs faces and license plates in the highway camera feeds to ensure strict public privacy.

Work Package 2: Building the AI Models

- Training the Vision Model: We will train a neural network to look at highway photos and classify the physical road surface as dry, wet, snowy, or icy.
- Scoring Alert Severity: We will build a text filter that reads the clean emergency alerts and assigns them a safety score based on the severity of the vocabulary.
- Combining the Data: We will develop a fusion algorithm that mathematically blends the camera results, alert scores, and weather telemetry to calculate a final safety percentage.
- Tuning for Safety: We will tune our model's settings to aggressively prioritize catching severe safety hazards, ensuring it warns drivers about danger rather than just optimizing for the shortest route.

Work Package 3: Setting Up the Backend

- Standardizing Workspaces: We will create a shared software requirements list, so every team member's computer runs the project the same way without dependency crashes using GitHub.
- Setting Up Automation: We will write background scripts that allow the system to run on its own, automatically fetching new data and updating predictions every few minutes.
- Saving Model Work: We will export our finished AI models into lightweight files so the backend can load them instantly during live operations.
- Packing into Containers: We will package the entire application into containers, acting as portable software boxes that let the system run smoothly on any local machine.

Work Package 4: Designing the App Interface

- Sketching the Layout: We will design the user interface around the Google Maps API, deciding exactly where the safety warnings and route cards will appear for the driver.
- Building the Dash Dashboard: We will use the Dash library to build an interactive dashboard, including a custom slider that lets users evaluate safety scores across a specific range of coordinates.
- Adding the Map Overlay: We will integrate a custom map layer that displays two to three route suggestions based on the user's origin, destination, and our live safety scores.

Q3: Task Owners

Assign roles and responsibilities for each team member. In other words, assign a task owner for each task within the WPs. This must be done strategically based on qualifications and interest of the team member being assigned a task. Common roles include project manager, data engineer, AI/ML engineer, MLOps engineer, etc. Divide the workload equally among the team members so that it is fair, balanced, and realistic. In the space below, list the tasks from Q2, and indicate the name of the team member that will work on the specific task in parenthesis. Also, provide a specific description (in 1 to 2 sentences) why this specific team member is appropriate to work on the assigned task. [1 mark]

Evans is Backend Engineer:

- Evans is interested in the backend development, and he has experience building a website for his book recommendation system.
- Configures the implementation environment.
- Manage the data that is used to communicate with the backend and frontend together with Xian.
- Build the web application and integrate map features for the user interface.

Xian is Data Engineer:

- Xian has already found some useful datasets while discussing projects, and he has lots of thoughts on adding more features to improve the idea; this job is consistent with the work he has already done.
- Search the historical data that is used to train models and the possible live API data that is for prediction.
- Support the feature engineering work with Yanan to train the model.
- Help to set up the data interaction between the frontend and the backend, and build the containers.

Yanan is MLE:

- Yanan has done some research on the models, and this is her dream job.
- Research the peer-reviewed papers and select 2 to 3 models that are good for the experiments.
- Work with Xian to preprocess the data, decide the features, train the model, and help to deploy the best model.
- Maintain the structure of the system, ensure the design is updated visually, and publish to the team on time if any updates arise.

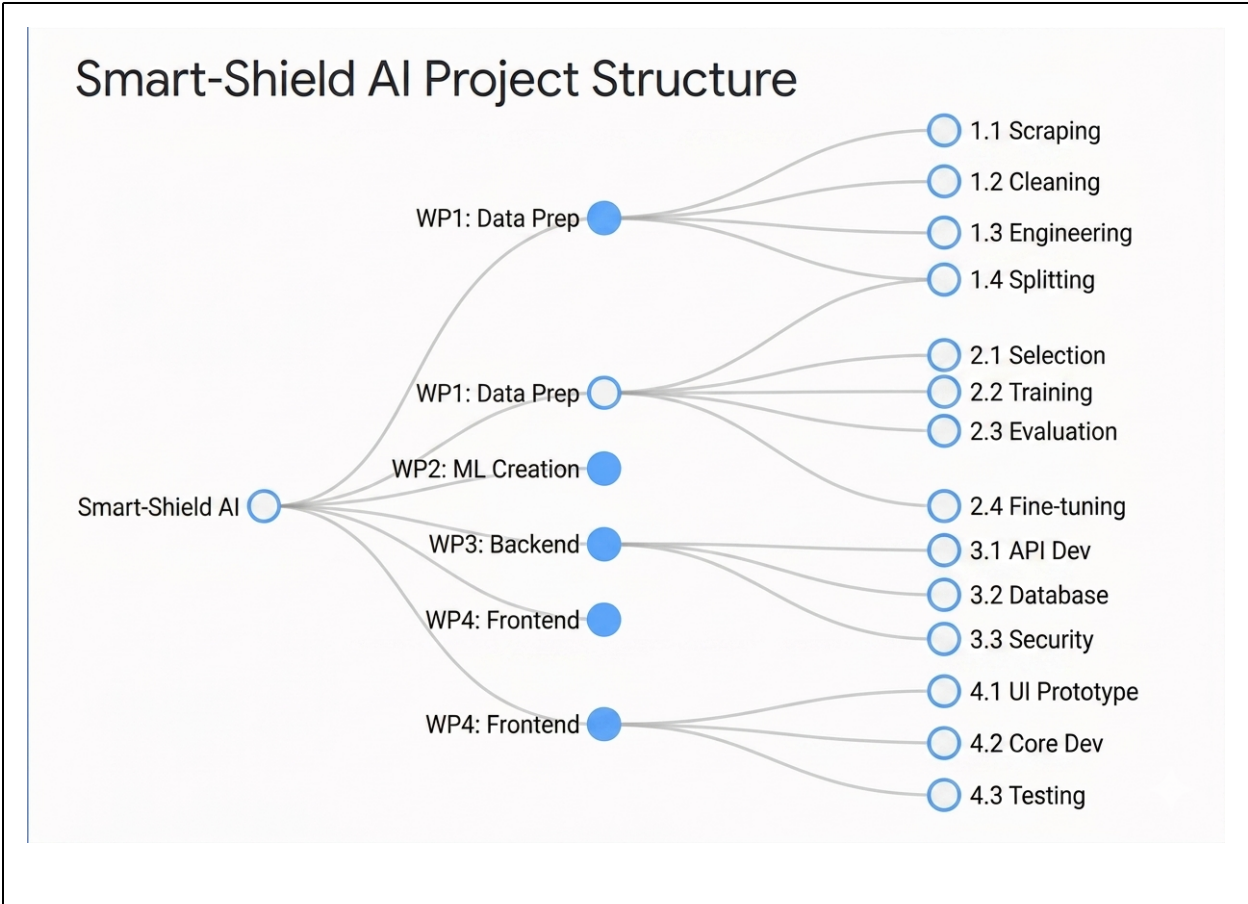
Afolabi is AI Optimizer:

- Afolabi is the team lead, and he has done some research on optimization of the scoring function.
- Help to test and refine the model and user experience.
- Implement the model together with Yanan, and optimize the scoring function to support the robust score system.
- Communicate with other teams and staff, support the hardware requirements, and help design and develop the frontend with Evans.

All team members will test the demo and help to prepare the final documents.

Q4: WBS Diagram

Using the descriptions in Q1 (work packages), Q2 (tasks), and Q3 (task owner) organize the WBS to visually capture relevant information by way of a tree diagram. The diagram should show the hierarchical relationships among all WPs and tasks, and it should include all relevant sub branches. Also, make the diagram complete and coherent by enumerating the topics and by placing arrows to show all interconnected associations. The diagram should visually display an overall summary for the workflow for the AI/ML project. Specialized software can be used to make the diagram (e.g., Visio, draw.io, photoshop, etc.). [1 mark]



Q5: Weekly Milestones

Define weekly milestones and deliverables for project tracking. The weekly milestones plan should cover Week 5 to Week 14 of the semester. Each week, a certain aspect of the project work needs to be completed (e.g., data gathering and cleaning, AI/ML modeling, model optimization, testing and prototyping, deployment and UI, tech report writing, final presentation and demonstration). In the space below, enumerate the weeks from Week 5 to Week 14, and provide a bulleted list of the tasks and sub tasks that the group aims to realistically accomplish by the end of each week. [2 marks]

Week	Task	Subtasks & Deliverables
5	Data & API Setup	- Find valid open source Toronto/Ontario traffic collision datasets. - Configure Google Map API to get route coordinates. - Setup Git repo and project environments.

Week	Task	Subtasks & Deliverables
6	Data Pipeline Ingestion	• Create scripts to fetch live weather and road conditions. • Make automated scrapers for public traffic camera data. • Clear missing values, standardize text alerts and structured data.
7	Computer Vision (CV)	• Build pipeline to process surveillance photos and video frames. • Classify road surface hazard. • Prototype a Neural Network for hazard recognition.
8	NLP & Baseline ML	• Build pipeline to extract hazard keywords from text alerts. • Train baseline models. • Evaluate models using safety focused metrics instead of general accuracy.
9	Multimodal Data Fusion	• Combine CV outputs and NLP text insights with live weather data. • Train final model to compute segment risk scores. • Resolve data latency to ensure real time capability.
10	Optimization & MLOps	• Optimize algorithms to reduce required local/cloud computing power. • Put data and modeling pipelines into Docker containers. • Deploy asynchronous background environment to precompute risk scores.
11	Google Maps UI Setup	• Develop front end web application. • Embed standard Google Maps interface with inputs. • Design backend API endpoints to connect UI with Dockerized ML models.
12	System Integration	• Display routes color coded by safety scores. • Test end to end functionality of the Google Maps web extension. • Run performance test to eliminate data rendering and latency.
13	Validation & Report	• Test the live environment under simulated high risk conditions. • Compile final model metrics and stack choices. • Clean code and perform final merges into the main branch.

Week	Task	Subtasks & Deliverables
14	Final Demonstration	- Finalize presentation slides and project live demo. - Run Dockerized background services for the live Google Maps. - Submit the final results.

Q6: Gantt Chart

Using the results in Q4 (WBS diagram) and Q5 (weekly milestones), develop a detailed Gantt chart. The Gantt chart is a project management tool that visually illustrates a project schedule over time. The Gantt chart should display a list of WPs and tasks vertically while being mapped to specific task owners. A timeline is shown horizontally using horizontal bars that should be used to represent the needed duration in weeks. Make the Gantt chart as detailed as possible. Also, important AI/ML milestones should explicitly be indicated on the Gantt chart. An example of a Gantt chart that can be replicated is shown in Appendix A. [2 marks]

WBS	Task Description	Owner	Start	Due	W5	W6	W7	W8	W9	W10	W11	W12	W13	W14
WP1	Data Gathering & Prep	Evans / Xian	W5	W6	■	■								
1.1	API & Live Data	Evans / Xian	W5	W5	■	■								
1.2	Historical Data	Evans	W5	W6	■	■								
1.3	Vision Pre-Processing	Yanan	W6	W6	■	■								
1.4	Text Pre-Processing	Yanan	W6	W6	■	■								
WP2	ML Model Creation	Yanan / Afolabi	W7	W10			■	■	■	■				
2.1	Vision Brain (CNN)	Yanan / Afolabi	W7	W8			■	■						
2.2	NLP Brain Dev	Yanan / Afolabi	W7	W8			■	■						
2.3	Fusion Optimizer	Afolabi	W9	W9					■	■				
2.4	Hyperparameter Tuning	Afolabi	W10	W10					■	■				
WP3	Backend System	Xian / Evans/Yanan	W9	W12					■	■	■	■		
3.1	Env Configuration	Evans	W5	W5	■	■								
3.2	Async Automation	Xian	W9	W10					■	■				
3.3	Containerization	Xian	W11	W12							■	■		
WP4	Frontend Application	Evans	W11	W14							■	■	■	■
4.1	UI/UX Wireframing	Evans	W11	W11							■	■		
4.2	HTML, CSS, Javascript	Evans	W11	W12							■	■		
4.3	Map Integration	Evans	W12	W12							■	■		
QA	Testing & Polish	Team	W13	W14									■	■

Q7: Version Control

Establish version control practices and repository management protocols with Git/GitHub. This tool will allow tracking changes in source code and related files over the span of the AI/ML capstone project. This tool will also allow practical collaboration with team members without overwriting each other's work. In the space below, explain how Git/GitHub will be used for version control and for repository planning. Also describe in full detail the collaboration workflow and the strategy that will be applied. [1 mark]

1. Repo Planning and Layout

To keep the project clean and clear, the repository remains modular. Data engineering, modeling, and application coding are separated.

```
|-- .gitignore
|-- README.md
|-- requirements.txt
|-- data/
|   |-- raw/
|   |-- processed/
|-- notebooks/
|-- src/
|   |-- data_pipeline/
|   |-- cv_pipeline/
|   |-- nlp_pipeline/
|   |-- models/
|   |-- app/
deployment/
```

- **.gitignore:** Raw data, temporary local files, environment variables, and heavy IDE configurations are excluded.
- **Data and Model Tracking:** Git LFS or DVC manages large datasets and trained models. Only lightweight pointer files are stored in the GitHub repository.

2. Collaboration

Direct commits to the main branch are strictly prohibited. The following steps are required.

1. **Isolate:** Before starting work, pull from the repository to get the latest version. Create a local branch using the code `git checkout -b feature/data-preprocessing`.
2. **Commit Often:** Make atomic and logical commits. Commit messages must follow a standard format. Use examples like `feat: add MinMax scaling to data pipeline` or `fix: resolve convergence warning in KNN training`.
3. **Push and Align:** Push the local branch to GitHub to make it visible to the team. Use the code `git push origin feature/data-preprocessing`.

3. Branch and Merge

Naming Conventions:

- `feature/cv-hazard-detection`: CNN and OpenCV pipeline development.
- `feature/nlp-alert-parser`: NLTK text processing for emergency alerts.
- `feature/data-ontario511-api`: Data layer ingestion from APIs.
- `feature/mlops-docker-bg`: Docker environment for background risk precomputation.
- `feature/web-plugin-ui`: Frontend JavaScript map ranking implementation.

4. Merging and Conflict Mitigation

Pull Requests and Code Review:

- **Interface Review:** Any pull request altering a pipeline output requires review by at least one other team member.
- **Squash and Merge:** Upon approval, branches are squashed into a single clean commit on the main branch.

Jupyter Notebooks:

- **Output Clearing:** All cell outputs must be cleared prior to committing.
- **Script Refactoring:** Once a model or function is validated, it must be refactored into a Python script inside the `src/` folder.

Q8: Limitations, Constraints, and Risks

Identify potential limitations, constraints, and risks that may be faced during this AI/ML capstone project. Keep in mind that financial support from Sheridan to teams is not available, and that access to GPU supported labs or clusters is not guaranteed. In the space below, develop a risk mitigation table that highlights the top 5 to 7 risks that many encounter while working on this project. The table should also include a contingency plan for each risk. The contingency should propose a backup strategy to mitigate the potential risk, to avoid disruptions to the operations of the workflow, and to allow completion of the AI/ML project. In the table below, provide a heading for each cell of the table that summarizes the risk and the related contingency. Then, provide a description for each cell of the table (in 2 to 3 sentences). [2 marks]

No.	Limitations, Constraints, Risks	Contingency Plan
#1	Weather Unpredictability Weather unpredictability can lead to inaccurate real time predictions during the testing phase.	Mock Data Representation This challenge can be solved by providing a mock data representation of extreme weather.
#2	Time Constraints Time constraints make it difficult to achieve all goals that have been set.	Realistic Goal Setting and Scheduling Realistic goals and proper scheduling of tasks and time allocations will be set to maximize productivity and ensure all goals are met.
#3	Lack of Project Budget The project lacks a financial budget.	Utilization of Free Services Free API services and other available internet tools will be utilized, and necessary funds will be pooled if required.

No.	Limitations, Constraints, Risks	Contingency Plan
#4	Hardware Constraints There are hardware constraints such as the lack of a dedicated GPU.	Cloud Service Integration Google Colab will provide cloud services that make processing power available over the internet while using local CPUs.
#5	Data Format Misalignment There is a risk of data format misalignment across different datasets.	Standardized Mapping Class Custom translation dictionaries in Pandas will be built to automatically clean up and align different names and coordinate systems into a single format.
#6	Frontend Map Lag and Latency Processing dense route data on an interactive map can cause the web browser user interface to lag.	Route Coordinate Downsampling Coordinate reduction techniques will be used to simplify long route lines, keeping the map fast and responsive while maintaining clear safety color coding.

Q9: Project Outcomes

By Week 14, all teams must prepare: (i) D4, a formal technical report detailing the comprehensive work on the AI/ML capstone project, (ii) D5a, slide presentation of the technical report, and (iii) D5b, a working live demonstration of the AI/ML solution. Q9.1 What is the anticipated major outcome of the capstone project? Q9.2 What type of AI/ML method will the solution consider (i.e., regression, classification, clustering, or possibly a combination of different methods)? Q9.3 What type of UI/system will showcase a working demonstration of the AI/ML solution (e.g., MobileApp, cloud service, locally or publicly hosted URL, AI powered physical system like IoT, drone, or robot, or possibly a combination of various UI/systems)? In the space below, provide specific answers to these three questions to demonstrate that the anticipated capstone results are thoroughly considered and strategically planned. [2 marks]

9.1 Expected Outcome The outcome is a web application. Once the user gives the start and destination, the website shows the safety index or accident risk for the real route recommended by Google Maps.

9.2 Methods The DNN, RandomForest, or other machine learning models that are recommended by the peer-review paper are used to train and predict the accident rates. And the TF-IDF or other NLP techniques to handle the real-time text data from 511 websites and weather data. Logistic regression model with SAGA solver to calculate the final safety scores.

9.3 UI & System The website will be on a local host for a live demo. It allows users to input the start location and destination, and the map will show the recommended routes, with the safety scores displayed. This is an interactive map. It will show the

colored routes with the predictions from the model. There is no need to install local software.

References

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